CLOSED-LOOP DC–DC BUCK CONVERTER

LTspice Simulation and Performance Report

LTspice 24.1.1 | Simulation-based power-electronics design | 9 October 2026

Abstract

This project develops and evaluates a closed-loop step-down (buck) DC–DC converter in LTspice. The goal is to convert a nominal 12 V DC input into a regulated output near 5 V using a 1 MHz switching command. The model combines a controlled switch, freewheel diode, 22 µH inductor, 100 µF output capacitor, resistive load, and behavioural feedback control. Simulations examine start-up, steady-state ripple, load transient response, input-voltage variation, switching behaviour, and duty-cycle adjustment. This is a simulation prototype and has not been validated on physical hardware.

1. Project overview

A buck converter reduces a DC input voltage to a lower DC output voltage. This is useful in embedded systems, instrumentation, communications and other electronics that need a stable low-voltage rail. The design target is approximately 5 V from a nominal 12 V input. The model is inspired by practical power-conversion applications but is not a space-qualified or hardware-validated design.

2. Main circuit schematic

The schematic below is the main reference for the report. On the left is the input source and controlled switch. The diode provides a freewheel path when the switch turns off. L1 and C1 form the output filter, with R1 as the load. The right side contains the duty command, repeating ramp, PWM behavioural source, and bounded integrator. The input is stepped from 12 V to 10 V at 3 ms, while the load is changed from 10 Ω to 5 Ω at 2 ms.

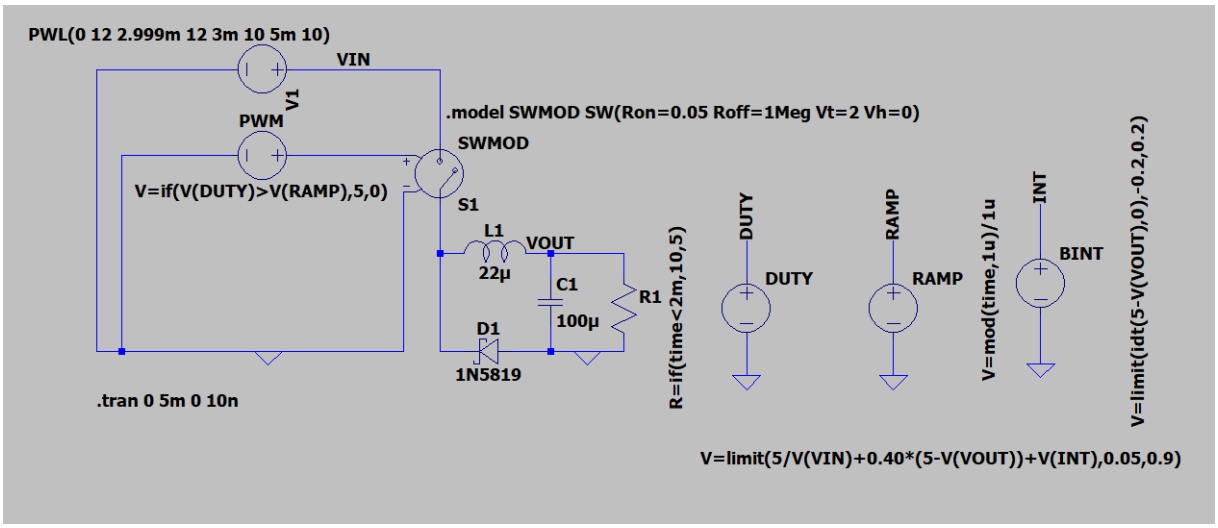


Figure 1. Main LTspice schematic: power stage, load, PWM generation, feedback controller, input-step source and transient command.

3. How the simulation was executed

Step 1 — Build the power stage

The circuit was assembled with a controlled switch S1, diode D1 (1N5819), inductor L1 = 22 µH, capacitor C1 = 100 µF and resistive load R1. The switch model uses Ron = 0.05 Ω, Roff = 1 MΩ, Vt = 2 V and Vh = 0.

Step 2 — Set up the load transient

The load expression is R=if(time<2m,10,5), so the load changes from 10 Ω to 5 Ω at 2 ms. At an ideal 5 V output, this corresponds to a load current increase from about 0.5 A to 1 A.

Step 3 — Generate PWM

The ramp is V=mod(time,1u)/1u, which repeats every 1 µs (1 MHz). The comparator expression is V=if(V(DUTY)>V(RAMP),5,0), generating a 0–5 V pulse command according to the duty signal.

Step 4 — Apply feedback control

The duty command is V=limit(5/V(VIN)+0.40*(5-V(VOUT))+V(INT),0.05,0.9). The feedforward term approximates the ideal duty ratio. The proportional term responds to the instantaneous output error, while V(INT)=limit(idt(5-V(VOUT),0),-0.2,0.2) provides bounded integral action.

Step 5 — Apply the input-voltage disturbance

The input source is PWL(0 12 2.999m 12 3m 10 5m 10), which holds the input at 12 V until 3 ms and then reduces it to 10 V.

Step 6 — Run transient analysis

The directive .tran 0 5m 0 10n simulates the first 5 ms with a maximum timestep of 10 ns. The waveform viewer is used at both full-run and zoomed-in time scales.

4. Circuit operation

When S1 is on, energy is delivered from the input to the inductor and load, and inductor current rises. When S1 turns off, inductor current continues through the freewheel diode. The output capacitor smooths the switched energy into a near-DC voltage. For an ideal buck converter, VOUT ≈ D × VIN: the ideal duty ratio is about 0.417 at 12 V input and 0.5 at 10 V input for a 5 V output. The feedback controller adjusts duty to compensate for output error and input/load disturbances.

5. Graph-by-graph explanation

Figure 2 — Full transient response: VIN (green) and VOUT (red)

The input remains near 12 V until 3 ms, then steps down to 10 V. The output starts high and settles near 4.95 V, with a disturbance around the 2 ms load step. This view shows overall regulation, but the shared voltage axis makes small output ripple hard to see.

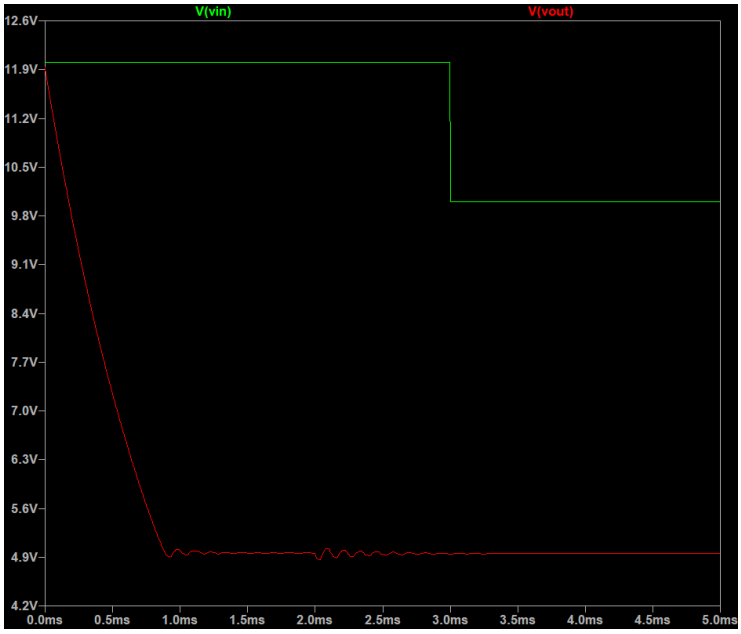


Figure 2. Full transient response: VIN (green) and VOUT (red), 0–5 ms.

Figure 3 — Steady-state VOUT ripple

The zoomed output trace stays around 4.95 V. The reported measurement log gives average output 4.9500028 V and ripple of 2.399 mV peak-to-peak in this window. This is a simulated result and depends on the idealised component models.

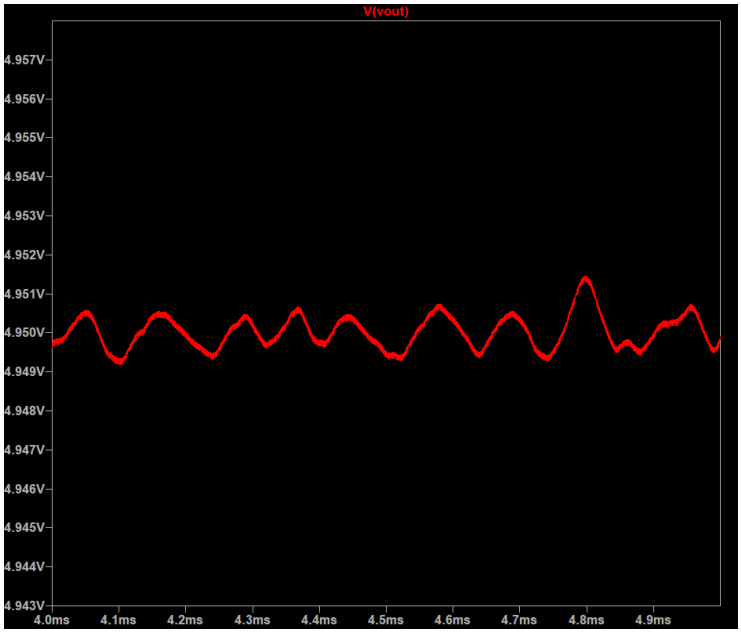


Figure 3. Steady-state VOUT ripple, 4.0–4.9 ms.

Figure 4 — Repetitive switching waveform at node V(n001)

The trace toggles between approximately 0 V and 5 V. The pulse spacing is consistent with a period near 1 µs, or about 1 MHz. Since the node is labelled V(n001), its exact role should be verified against the schematic before identifying it definitively as the switch-drive signal.

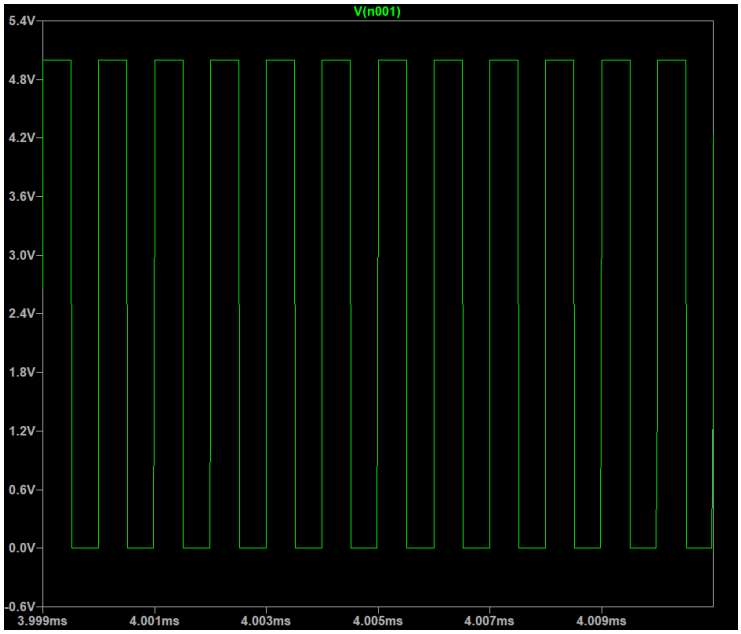


Figure 4. Repetitive switching waveform at node V(n001), approximately 4.0–4.01 ms.

Figure 5 — VOUT response around the load step at 2 ms.

The load changes from 10 Ω to 5 Ω at 2 ms. The reported output minimum in the 2–2.5 ms window is 4.8660 V and the maximum is 5.0227 V. The damped oscillation shows transient undershoot and overshoot before recovery.

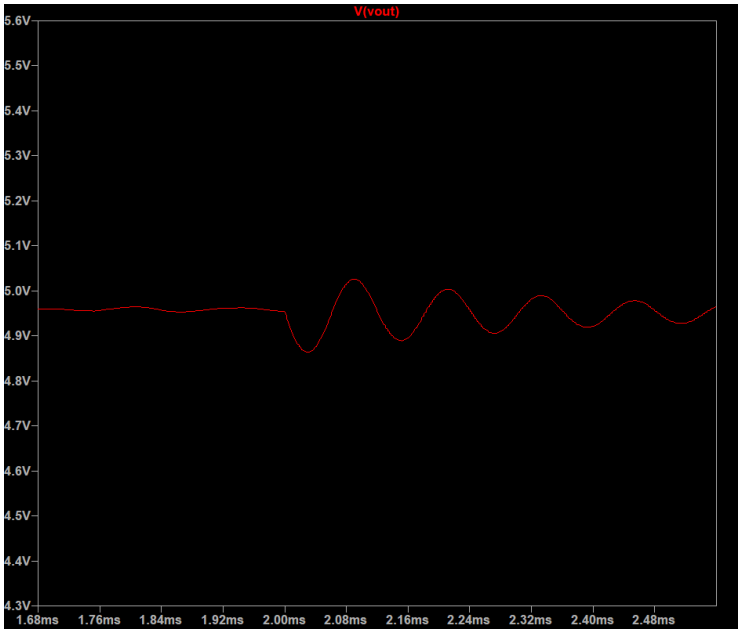


Figure 5. VOUT response around the load step at 2 ms.

Figure 6 — Duty command V(DUTY)

The duty command settles near 0.43 before the input step and increases to roughly 0.518 after VIN drops to 10 V. This direction is expected because a lower input requires a higher duty ratio to maintain the same output voltage.

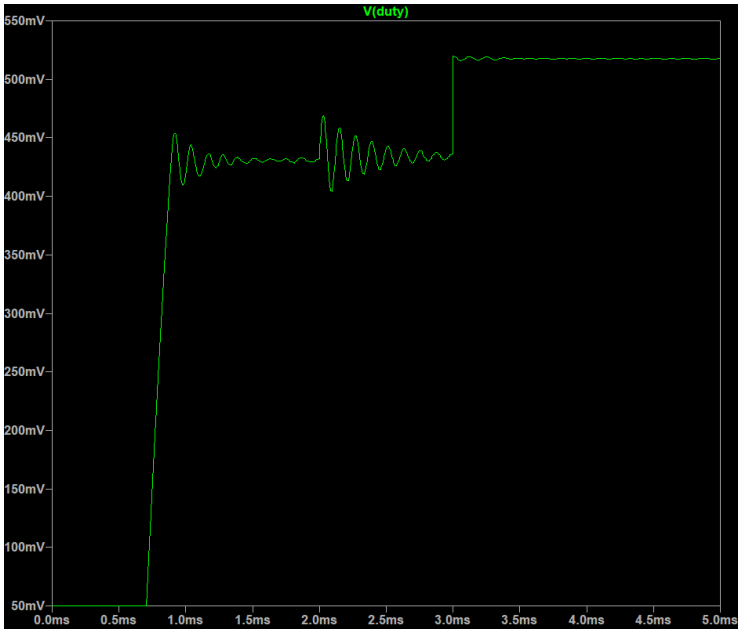


Figure 6. Duty command V(DUTY), 0–5 ms.

6. Summary of results

Metric	Result	Meaning
Average VOUT, 4.0–4.9 ms	4.9500028 V	Approximately 5 V; at the lower edge of a ±1% band.
Steady-state VOUT ripple	2.399 mV p-p	Below the project target of 100 mV p-p in this simulation.
Load-step minimum	4.8660 V	Undershoot after load increases.
Load-step maximum	5.0227 V	Overshoot during transient recovery.
Average duty command	0.517569 (51.76%)	Consistent with the reduced 10 V input, allowing for feedback correction.

Metric	Result	Meaning
Inductor current plot	Not supplied	I(L1) graph is still recommended to document current ripple.

7. Limitations and recommended next steps

• This report documents simulation only; no PCB or bench validation is evidenced.

• Efficiency has not been measured from input and output power.

• The full 9–14 V input range has not been tested; only 12 V and 10 V are shown.

• Add I(L1) to show inductor-current ripple and assess continuous-conduction behaviour.

• Consider realistic inductor DCR, capacitor ESR, diode/switch losses and layout parasitics.

• Formal loop stability margins and hardware protection have not been established.

8. Conclusion

The LTspice model demonstrates a behavioural closed-loop buck converter with a nominal 5 V output, 1 MHz PWM command, input-voltage step and load transient. The plots show an average output near 4.95 V and a reported steady-state ripple of about 2.4 mV peak-to-peak. The response to the load step and the duty-cycle increase after the input step are visible in the supplied plots. The project is suitable to present as a simulation prototype, with further validation required before claiming hardware readiness.

Appendix — Key LTspice expressions

• Input: PWL(0 12 2.999m 12 3m 10 5m 10)

• Load: R=if(time<2m,10,5)

• Ramp: V=mod(time,1u)/1u

• PWM: V=if(V(DUTY)>V(RAMP),5,0)

• Integrator: V=limit(idt(5-V(VOUT),0),-0.2,0.2)

• Duty: V=limit(5/V(VIN)+0.40*(5-V(VOUT))+V(INT),0.05,0.9)

• Transient: .tran 0 5m 0 10n

Supplement: Inductor-Current Waveform

Figure 7 shows I(L1), the current through the 22 µH inductor over the 0–5 ms transient simulation. The current rises during start-up and increases again around 2 ms when the load changes from 10 Ω to 5 Ω. A large transient peak and damped oscillation follow, after which the current settles around approximately 1 A. This is consistent with the higher load current after the load step. Because this is a full-run view, individual 1 MHz ripple cycles are not resolved clearly; a zoomed-in plot over approximately 4.0–4.01 ms would show the switching ripple more clearly.

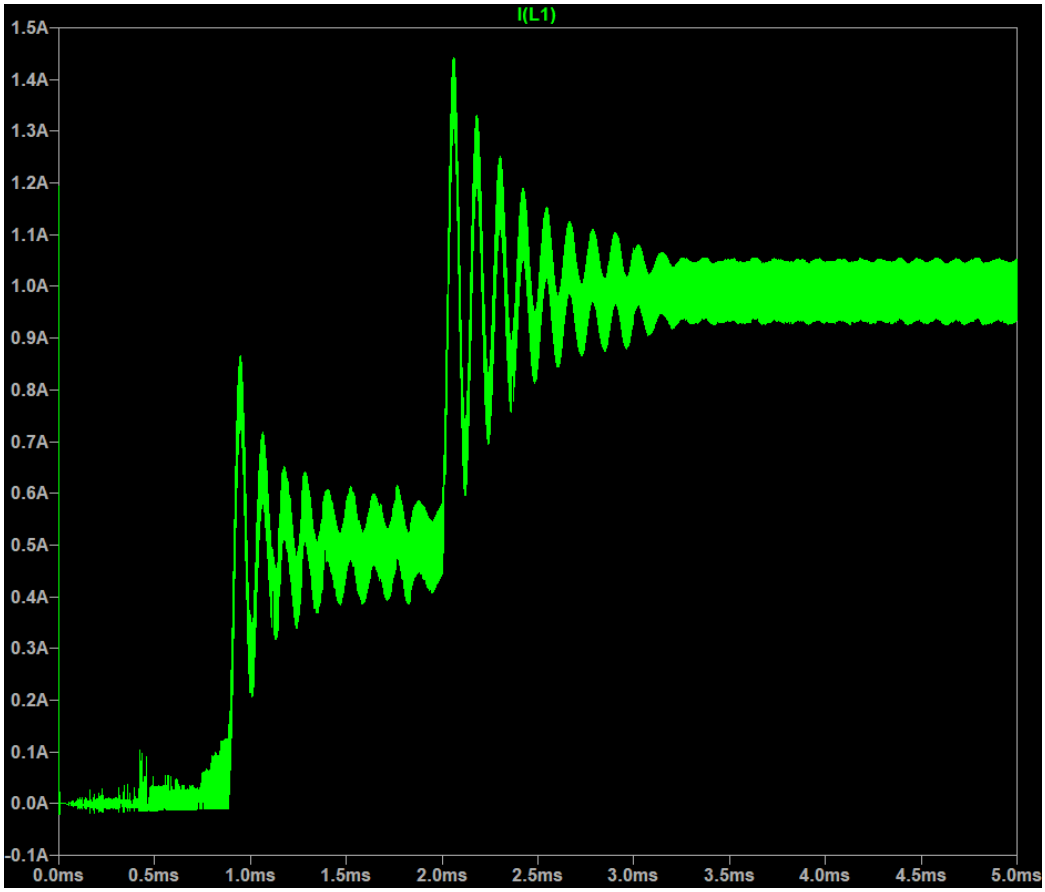


Figure 7. Inductor current I(L1) from 0 to 5 ms, showing start-up, the load-step transient, and steady-state current.

Figure 8 — Zoomed inductor-current ripple at steady state

This zoomed I(L1) plot shows the inductor current over approximately 3.996–4.011 ms. The waveform has a repeating triangular pattern consistent with the 1 MHz switching period. From the displayed scale, current varies approximately from 0.93 A to 1.05 A, giving an estimated peak-to-peak ripple of about 0.12 A. This visual estimate is close to the earlier measurement-log value of approximately 0.130 A peak-to-peak; the exact value should be taken from LTspice cursor or .meas results rather than estimated from the screenshot.

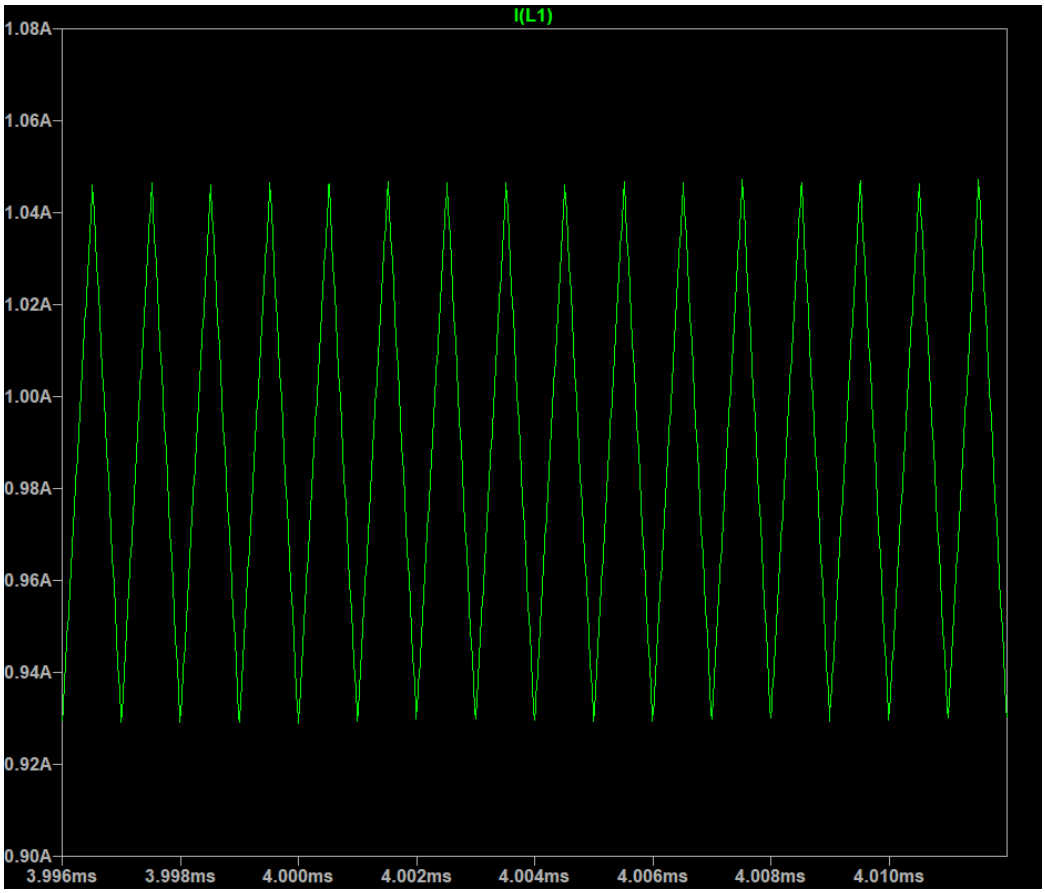


Figure 8. Zoomed inductor current I(L1), showing switching-cycle ripple near 4 ms.